

TABLE OF CONTENTS

Title	Page No.
ABSTRACT	i
ACKNOWLEDGEMENT	iii
TABLE OF CONTENTS	v
LIST OF TABLES	viii
LIST OF FIGURES	ix
LIST OF ABBREVIATIONS	xii
CHAPTER 1 INTRODUCTION	1
1.1 MOTIVATION AND MAJOR OBJECTIVES	2
1.2 CONTRIBUTIONS	3
1.3 THESIS ORGANIZATION	4
CHAPTER 2 LITERATURE STUDY	7
2.1 WATER QUALITY MONITORING IN AQUACULTURE	7
2.2 WATER QUALITY PREDICTION IN AQUACULTURE	11
2.3 DATASETS	20
2.3.1 ADAK water quality dataset	20
2.3.2 MAC water quality dataset	20
2.4 PERFORMANCE METRICS	20
2.5 Loss Function	21
CHAPTER 3 DEEP LEARNING BASED VIRTUAL SENSING FOR AFFORDABLE WATER QUALITY MONITORING IN AQUACULTURE	23
3.1 INTRODUCTION	23
3.2 PROPOSED METHODOLOGY OF AFFORDABLE WATER QUALITY MONITORING IN AQUACULTURE	25
3.2.1 Virtual sensing based water quality monitoring	25
3.2.2 Deep learning enhanced virtual sensing for water quality mon- itoring	26

3.3	EXPERIMENTAL SETUP FOR TESTING THE PROPOSED METHODS	27
3.3.1	Hardware	28
3.3.2	Software	29
3.4	RESULTS AND DISCUSSIONS	30
3.4.1	ADAK water quality dataset	30
3.4.2	MAC water quality dataset	31
3.5	SUMMARY	32
 CHAPTER 4 DEEP LEARNING MODELS FOR WATER QUALITY PREDICTION IN AQUACULTURE		33
4.1	INTRODUCTION	33
4.2	DATA COLLECTION AND ANALYSIS	35
4.2.1	Acquisition of data	35
4.2.2	Correlation analysis	37
4.2.3	Data pre-processing	37
4.3	DEEP LEARNING MODELS	38
4.3.1	LSTM DL-RNN	38
4.3.2	GRU DL-RNN	39
4.3.3	Bidirectional LSTM/GRU DL-RNN	40
4.3.4	Model implementation	41
4.4	RESULTS AND DISCUSSIONS	41
4.4.1	ADAK water quality dataset	41
4.4.2	MAC water quality dataset	42
4.5	SUMMARY	44
 CHAPTER 5 HYBRID DEEP LEARNING MODELS FOR ACCURATE AND COMPUTATIONALLY EFFICIENT WATER QUALITY PREDICTION IN AQUACULTURE		47
5.1	INTRODUCTION	47
5.2	PROPOSED HYBRID CNN-LSTM/GRU DL-RNN MODELS	49
5.2.1	Model implementation	51
5.3	EXPERIMENTS: HYPERPARAMETER TUNING	51
5.3.1	Learning rate (L_r)	52
5.3.2	Epoch (E_p)	52
5.3.3	Window size (W_z)	55
5.3.4	Batch size (B_z)	55
5.3.5	Computation time	59
5.4	RESULTS AND DISCUSSIONS	59
5.4.1	ADAK water quality dataset	60

5.4.2	MAC water quality dataset	61
5.5	SUMMARY	63
CHAPTER 6 ATTENTION MECHANISM BASED DEEP LEARNING MODELS FOR IMPROVING THE ACCURACY OF WATER QUALITY PREDICTION IN AQUACULTURE . .		65
6.1	INTRODUCTION	65
6.2	PROPOSED ABD-RNN MODELS	67
6.2.1	Bidirectional LSTM/GRU DL-RNN layer	67
6.2.2	Attention layer	68
6.2.3	Model implementation	70
6.3	EXPERIMENTS: HYPERPARAMETER TUNING	70
6.3.1	Optimizers (O_p)	71
6.3.2	Learning rate (L_r)	71
6.3.3	Epoch (E_p)	71
6.3.4	Window size (W_z)	75
6.3.5	Batch size (B_z)	77
6.3.6	Computation time	77
6.4	RESULTS AND DISCUSSIONS	77
6.4.1	ADAK water quality dataset	80
6.4.2	MAC water quality dataset	84
6.5	SUMMARY	86
CHAPTER 7 CONCLUSIONS		87
7.1	SUMMARY OF THESIS	87
7.2	FUTURE WORK	88
REFERENCES		89
LIST OF PAPERS BASED ON THESIS		99
BRIEF CURRICULUM VITAE		103
DOCTORAL COMMITTEE		105